PROJECT SYNOPSIS: SMART EV CHARGING SCHEDULE OPTIMIZATION

IoT-Enabled Mathematical Optimization using Synthetic & Real ESP32 Sensor Telemetry with HiveMQ MQTT & Web Control Deck

Project Domain	IoT, Smart Grid, Energy Optimization & Cloud Telemetry
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Assessment Type	Center of Excellence (COE) Project Evaluation
Hardware Microcontroller	ESP32 DevKit V1 (Xtensa Dual-Core 32-bit, 240MHz)
Sensors & Actuators	ACS712-30A Current Sensor, ZMPT101B AC Voltage Sensor, DHT22 Temperature, 30A Relay Contactor
Cloud Message Broker	HiveMQ MQTT Broker (broker.hivemq.com:1883 & Secure WSS:8884)
Logic & Optimization Stack	SciPy Linear Programming (HiGHS Solver), Node-RED Dashboard, Streamlit, HTML5 / Tailwind Dashboard
Deployment Status	Live on GitHub Pages (https://gsparsh580-coder.github.io/smart-ev-charging-optimization/)

1. Introduction & Background

The accelerated global adoption of electric vehicles (EVs) introduces substantial localized electrical load on residential power distribution grids. When commuters return home in the evening (17:00–21:00) and plug in their vehicles concurrently, uncoordinated charging causes severe peak load spikes that threaten distribution transformer reliability and force utilities to fire carbon-intensive peaker power plants. Furthermore, residential electricity tariffs during these evening hours are typically set at their maximum peak rates under Time-of-Use (TOU) billing schemes. This project designs, implements, and deploys a complete end-to-end Smart EV Charging Schedule Optimization Platform that utilizes mathematical optimization (Linear Programming) to minimize charging costs, shave transformer peak load, and maximize rooftop solar PV utilization while guaranteeing driver departure deadlines.

2. Problem Statement

Traditional residential EV charging exhibits three critical deficiencies:

• Coincident Peak Stress: Uncontrolled charging coincides exactly with peak domestic cooking, lighting, and air-conditioning loads, risking distribution transformer trip.

• High Energy Costs: Charging immediately upon arrival incurs peak electricity tariffs (INR 11.80/kWh), costing up to 2.5x more than off-peak charging.

• Lack of Solar Self-Consumption: Surplus daytime rooftop solar generation is often exported back to the grid at low feed-in tariffs rather than being used to charge the vehicle.

• Absence of Remote Control & Safety Monitoring: Standard EV chargers offer limited remote visibility, lacking real-time contactor thermal monitoring or emergency trip mechanisms.

3. Project Objectives

• Cost Minimization: Minimize total charging energy cost per session using mathematical Linear Programming.

• Departure Deadline Guarantee: Ensure the EV battery reaches the user-specified target State-of-Charge (SoC) prior to morning departure.

• Peak Load Shaving: Enforce household and transformer capacity ceilings ($P_{grid_limit} = 11.0$ kW) to prevent circuit breaker trips.

• Solar Self-Consumption Maximization: Schedule daytime charging during peak solar generation (10:00–15:00) to absorb green energy.

• Dual-Mode Telemetry Ingestion: Operate seamlessly via physical ESP32 sensors (ACS712, ZMPT101B, DHT22) or synthetic stochastic simulator.

• Serverless Web Control Deck: Provide a live zero-installation dashboard on GitHub Pages connected over HiveMQ WebSockets for universal remote access.

4. Mathematical Formulation & Optimization Algorithm

The scheduling problem is formulated as a Constrained Linear Program (LP) solved across 24 discrete one-hour time steps:

Minimize Sum_{t=1..24} [(C_t * P_t * dt) - (lambda_solar * min(P_t, S_t) * dt)]

Subject to constraints:

1. Battery State Dynamics: SoC_{t+1} = SoC_t + [(eta * P_t * dt) / E_capacity] * 100%

2. Departure SoC Guarantee: SoC(t_departure) >= SoC_target (e.g. 85% at 07:00 AM)

3. Charger Power Limits: 0 <= P_t <= P_max (e.g. 7.4 kW Level 2 AC) for active plug-in window; P_t = 0 when unplugged

4. Grid Capacity Ceiling: P_t + P_baseload,t <= P_grid_limit (11.0 kW)

Where C_t is the dynamic TOU tariff rate, P_t is charging power, S_t is solar PV output, eta = 0.92 is AC-to-DC conversion efficiency, and lambda_solar is the self-consumption incentive factor. Solved via scipy.optimize.linprog with high-speed HiGHS backend and client-side JavaScript fallback.

5. System Architecture & IoT Data Pipeline

The system integrates four tightly coupled architectural layers:

• Sensing & Actuation Layer: Physical ESP32 DevKit V1 interfacing with ACS712-30A current sensor (GPIO 34), ZMPT101B voltage transformer (GPIO 35), DHT22 temperature sensor (GPIO 14), and a 30A heavy contactor relay (GPIO 26).

• Messaging Layer: Cloud-hosted HiveMQ MQTT Broker (broker.hivemq.com) facilitating bidirectional asynchronous messaging over topics 'smart_ev/telemetry', 'smart_ev/control', 'smart_ev/schedule', and 'smart_ev/status'.

• Analytics & Optimization Engine: Python backend running SciPy optimization, multi-tariff modeling, and synthetic data generation.

• Supervisory Dashboard Layer: Serverless web dashboard hosted on GitHub Pages communicating via Secure WebSockets (port 8884), complemented by a Node-RED flow dashboard and Streamlit analytics.

6. Experimental Results & Financial Impact

Empirical validation across a standard residential EV session (60 kWh battery, 20% to 85% SoC target, 42.39 kWh energy delivered):

• Uncontrolled Charging Cost: INR 370.93 (charged immediately at peak 18:00–23:00 tariff of INR 11.80/kWh).

• Smart Optimized Cost: INR 148.37 (scheduled automatically during off-peak 00:00–06:00 tariff of INR 3.50/kWh).

• Monetary Savings: INR 222.56 saved per charge cycle (60.0% cost reduction).

• Carbon Footprint Reduction: 4.71 kg of CO2 avoided per session by shifting draw away from coal peaker plants.

• Annualized Impact: Assuming 250 annual charging cycles, household savings exceed INR 55,640/year with 1,177 kg of CO2 avoided.

• Peak Transformer Shaving: Coincident peak draw reduced from 7.4 kW to 0.0 kW during the 17:00–21:00 evening grid peak.

7. Hardware Bill of Materials (BOM)

Total prototyping hardware cost: INR 1,675. Key components:

• ESP32 DevKit V1 (30-pin, WROOM-32): INR 450

• ACS712-30A Current Sensor Module: INR 85

• ZMPT101B AC Voltage Sensor Module: INR 120

• DHT22 Temperature & Humidity Sensor: INR 95

• 30A 250V AC Heavy Contactor Relay Module (Optocoupled): INR 180

• HLK-PM01 5V 1A Regulated Power Supply: INR 220

• DIN Rail Screw Terminals, Enclosure, Breadboard & Passives: INR 525

8. Deployment & Verification

The project is publicly deployed and verified:

• GitHub Repository: https://github.com/gsparsh580-coder/smart-ev-charging-optimization

• Live Web Dashboard: https://gsparsh580-coder.github.io/smart-ev-charging-optimization/

• Direct broker connectivity enables immediate remote evaluation from any web browser worldwide without installing dependencies.

9. Conclusion

The Smart EV Charging Schedule Optimization Platform successfully validates that applying mathematical Linear Programming to smart grid IoT telemetry delivers measurable, high-impact economic and environmental benefits. By shifting 42.39 kWh of charging demand from peak to off-peak hours, the platform achieves a 60.0% cost reduction, eliminates transformer overloading, and provides a zero-install cloud interface that bridges physical IoT hardware with academic optimization theory.